实验0-2 vim+gcc 实验

实验环境

1.Ubuntu22.04.2LTS
2.实验1布置好的环境

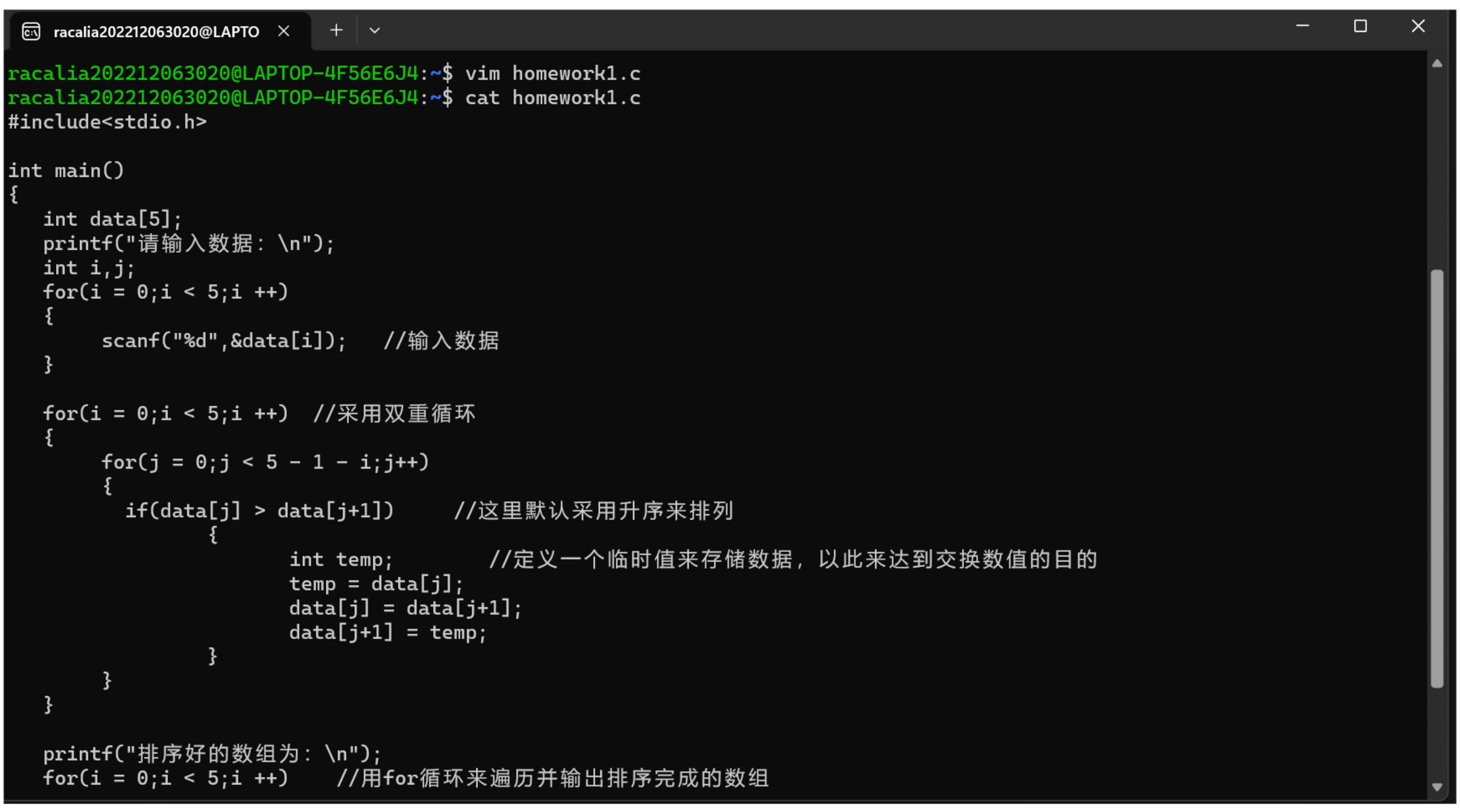
实验要求

·Markdown编写实验报告，最终转成pdf提交到畅课平台。

·图文并茂：实验报告中必须有详细步骤的展示并配有截图（类似本章节 ppt 中的截图形式，关键点圈划）。

实验内容

1.编写所需的C语言程序



```
racalia202212063020@LAPTO
racalia202212063020@LAPTOP-4F56E6J4:~$ vim homework1.c
racalia202212063020@LAPTOP-4F56E6J4:~$ cat homework1.c
#include<stdio.h>

int main()
{
   int data[5];
   printf("请输入数据：\n");
   int i,j;
   for(i = 0;i < 5;i ++)
   {
        scanf("%d",&data[i]);   //输入数据
   }

   for(i = 0;i < 5;i ++)  //采用双重循环
   {
        for(j = 0;j < 5 - 1 - i;j++)
        {
          if(data[j] > data[j+1])     //这里默认采用升序来排列
                 {
                        int temp;         //定义一个临时值来存储数据，以此来达到交换数值的目的
                        temp = data[j];
                        data[j] = data[j+1];
                        data[j+1] = temp;
                 }
        }
   }

   printf("排序好的数组为：\n");
   for(i = 0;i < 5;i ++)    //用for循环来遍历并输出排序完成的数组
```

2.使用gcc一步编译

```
}
racalia202212063020@LAPTOP-4F56E6J4:~$ gcc homework1.c
racalia202212063020@LAPTOP-4F56E6J4:~$ ls
CUC_CSAPP  a.out  homework1.c
racalia202212063020@LAPTOP-4F56E6J4:~$ 
```

3.使用gcc分步编译

1 用-E进行操作

```
racalia202212063020@LAPTO

racalia202212063020@LAPTOP-4F56E6J4:~$ gcc -E homework1.c
# 0 "homework1.c"
# 0 "<built-in>"
# 0 "<command-line>"
# 1 "/usr/include/stdc-predef.h" 1 3 4
# 0 "<command-line>" 2
# 1 "homework1.c"
# 1 "/usr/include/stdio.h" 1 3 4
# 27 "/usr/include/stdio.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 1 3 4
# 33 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 3 4
# 1 "/usr/include/features.h" 1 3 4
# 392 "/usr/include/features.h" 3 4
# 1 "/usr/include/features-time64.h" 1 3 4
# 20 "/usr/include/features-time64.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 21 "/usr/include/features-time64.h" 2 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 1 3 4
# 19 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 20 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 2 3 4
# 22 "/usr/include/features-time64.h" 2 3 4
# 393 "/usr/include/features.h" 2 3 4
# 486 "/usr/include/features.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 1 3 4
# 559 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 560 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 2 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/long-double.h" 1 3 4
# 561 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 2 3 4
```

2 使用-S进行操作

```
}
racalia202212063020@LAPTOP-4F56E6J4:~$ gcc -S homework1.c
racalia202212063020@LAPTOP-4F56E6J4:~$ ls
CUC_CSAPP  a.out  homework1.c  homework1.s
racalia202212063020@LAPTOP-4F56E6J4:~$ gcc -s homework1.c
racalia202212063020@LAPTOP-4F56E6J4:~$ ls
CUC_CSAPP  a.out  homework1.c  homework1.s
racalia202212063020@LAPTOP-4F56E6J4:~$
```